Explaining a classifier's decisions

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Abstract

Sed fringilla tempus hendrerit. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Etiam ut elit sit amet metus lobortis consequat sit amet in libero. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Phasellus vel sem magna. Nunc at convallis urna. isus ante. Pellentesque condimentum dui. Etiam sagittis purus non tellus tempor volutpat. Donec et dui non massa tristique adipiscing. Quisque vestibulum eros eu. Phasellus imperdiet, tortor vitae congue bibendum, felis enim sagittis lorem, et volutpat ante orci sagittis mi. Morbi rutrum laoreet semper. Morbi accumsan enim nec tortor consectetur non commodo nisi sollicitudin. Proin sollicitudin. Pellentesque eget orci eros. Fusce ultricies, tellus et pellentesque fringilla, ante massa luctus libero, quis tristique purus urna nec nibh.

Introduction

Logic-based Abductive Explanations

We consider a classifier $\kappa : F \to Cl$ where F is the feature space and Cl the set of classes. An instance $x \in F$ is described by a set of literals (f_i, v). A partial assignment $E \subseteq x$ is a subset of literals of x, and $E(y)$ means that y agrees with E on all features it defines.

CPI-Xp (Coverage-based Explanations)

Standard abductive explanations (AXp) implicitly assume **feature independence**. When dependencies exist (e.g. a pregnant person is necessarily a woman), AXp's can produce superfluous or redundant explanations, and their number can grow exponentially [**cooper˙abductive˙2024**].

Cooper and Amgoud [**cooper˙abductive˙2024**] address this by introducing **coverage-based explanations** (CPI-Xp). The key notion is the **coverage** of a partial assignment E:

$$cov(E) = \{x \in F[\mathcal{C}] \mid E(x)\} \tag{1}$$

where $F[\mathcal{C}]$ is the set of instances satisfying constraints $\mathcal{C}$. An explanation E' **subsumes** E if $cov(E) \subseteq cov(E')$.

Definition (CPI-Xp). A coverage-based PI-explanation of $\kappa(x)$ is a partial assignment E such that:

1. $E(x)$
2. $\forall y \in F[\mathcal{C}],\ E(y) \Rightarrow \kappa(y) = \kappa(x)$
3. No other valid explanation E' strictly subsumes E in $F[\mathcal{C}]$

By selecting explanations of **maximal coverage**, CPI-Xp automatically discards redundant ones. Two refinements exist:

- **mCPI-Xp**: subset-minimal among CPI-Xp's.
- **pCPI-Xp**: keeps only one representative among equivalent ones (same coverage). This is the only notion satisfying all seven desirable properties (success, non-triviality, irreducibility, coherence, consistency, independence, non-equivalence).

Complexity: Computing CPI-Xp over the full feature space is Π_2^P-complete. To achieve tractability, **dataset-based versions** (d-CPI-Xp) compute coverage over a sample $T \subseteq F[\mathcal{C}]$, reducing complexity to $O(m^2n^2)$, at the cost of explanations only valid for instances in T.

Shapley values based XAI

In this section, we present local explanation methods based on the Shapley values concept from game theory that give an importance score to each feature in the prediction of a single instance. Shapley values provide a mathematically fair and unique method to attribute the payoff of a cooperative game to the players. Formally, we define the coalitional game $(\mathcal{F}, v)$ composed of the set of the n numbered features used in the prediction $\mathcal{F} = \{1, \ldots, n\}$ and the evaluation function $v(\mathcal{S}) = \mathbb{E}[f(X)|X_{\mathcal{S}} = x_{\mathcal{S}}]$ for all sub-coalition $\mathcal{S} \subset \mathcal{F}$, where f is the classification function representing our model and $x_{\mathcal{S}} = y_{\mathcal{S}}$ means that instances x and y share the same values for the features in $\mathcal{S}$. The Shapley value for a feature $i \in \mathcal{F}$ is defined as follows:

$$\phi_i = \frac{1}{n} \sum_{\mathcal{S} \in \mathcal{F} \setminus \{i\}} \binom{n-1}{|\mathcal{S}|}^{-1} \left(\mathbb{E}[f(X)|X_{\mathcal{S}\setminus\{i\}} = x_{\mathcal{S}\setminus\{i\}}] - \mathbb{E}[f(X)|X_{\mathcal{S}} = x_{\mathcal{S}}]\right) \tag{2}$$

We see that the Shapley value of a feature i is computed by going through all the coalitions not containing i and looking at the payoff difference with and without the feature i involved.

From the efficiency axiom of the Shapley values, we get the following property:

$$\sum_{i \in \mathcal{F}} \phi_i(v) = \mathbb{E}[f(X)|X_{\mathcal{F}} = x_{\mathcal{F}}] - \mathbb{E}[f(X)|X_{\emptyset} = x_{\emptyset}] = f(x) - \mathbb{E}[f(X)] \tag{3}$$

Therefore, the sum of the importance is equal to the actual predicted value minus a reference contribution that is the mean of all the predictions for the distribution on the input X.

Figure 1 vizualises computed importance scores for the prediction of a naive Bayes classifier on an instance of the Titanic dataset.

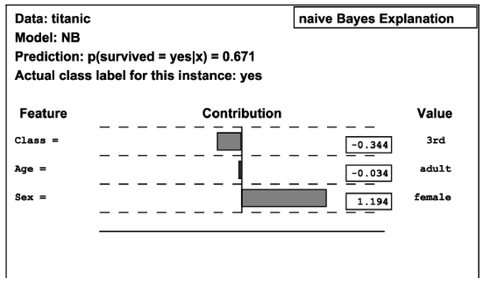

Figure 1: Feature importance attributions computed with Shapley values for the decision of a naive Bayes model on an instance of the Titanic dataset

Approximation

The computation of exact Shapley values is intractable as it requires to explore all the sub-coalitions which are in exponential number. Moreover, the computation of the evaluation function requires to explore all the input space. Previous work have come up with sampling methods to get an approximation in polynomial time. Sub-coalitions are sampled by taking all the features preceeding i in random orderings O_j of $\mathcal{F}$: $\mathcal{S}_j = Pre_i(O_j)$. The reference instances X are sampled following the distribution of input instances.

Perturbation model-agnostic method

Perturbation methods identify which input features most significantly influence a black-box model's decision process. These model-agnostic techniques compare the model's output for an original instance against its predictions for perturbed samples. By observing how these modifications affect the classification, we can quantify the importance of each feature without accessing the model's internal architecture.

LIME

LIME (Local Interpretable Model-Agnostic Explainer) generates local surrogates to approximate a black-box model f [**ribeiro˙why˙2016**].

Key Principle: It minimizes a loss function $\mathcal{L}(f, g, \pi_x)$ to ensure local fidelity while keeping the explanation model g simple enough ($\Omega(g)$) for humans.

Limitation: The explanation depends heavily on the kernel width π_x and sampling, leading to potential instability.

A most global explanation of the black-box model can be provided by SP-LIME which select a set of local and non-redundant LIME explanations.

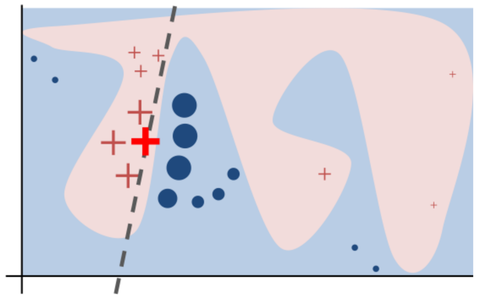

Figure 2: LIME linear local approximation. Black-box model f classes is represented in blue/gray. The red cross represents an explained instance and the others symbols are the perturbation samples generated from the original. The dotted line represents the local approximation of g.

Anchors

Anchors were developed to solve the lack of clear boundaries provided by LIME. To perform that, Anchors generates IF-THEN rules which 'anchor' a prediction. This set of short and disjoint rules are more comprehensible and then preferred by the users [**ribeiro˙anchors˙2018**].

Formal Definition: A rule A is an Anchor for x if $A(x) = 1$ and it maintains high precision τ among its neighbors:

$$\mathbb{E}_{\mathcal{D}(z|A)}[1_{f(x)=f(z)}] \geqslant \tau \tag{4}$$

with $\mathcal{D}(x \mid A)$ the distribution of neighbors of x wich satisfy the Anchor A.

Computing Anchors is a combinatorial optimization problem. To handle the exponential search space, a **Bottom-Up approach** is used:

- **Candidate Generation**: Rules are extended iteratively with additional predicates.
- **Selection via MAB**: A Multi-Armed Bandit (*KL-LUCB*) algorithm identifies the most precise candidates using minimal model calls [**ribeiro˙anchors˙2018**].
- **Beam Search**: To avoid purely greedy results, the algorithm maintains B candidates, finally outputting the one with the **highest coverage**: $cov(A) = \mathbb{E}_{\mathcal{D}(z)}[A(z)]$.

Conclusions

- Pellentesque eget orci eros. Fusce ultricies, tellus et pellentesque fringilla, ante massa luctus libero, quis tristique purus urna nec nibh. Phasellus fermentum rutrum elementum. Nam quis justo lectus.
- Vestibulum sem ante, hendrerit a gravida ac, blandit quis magna.
- Donec sem metus, facilisis at condimentum eget, vehicula ut massa. Morbi consequat, diam sed convallis tincidunt, arcu nunc.
- Nunc at convallis urna. isus ante. Pellentesque condimentum dui. Etiam sagittis purus non tellus tempor volutpat. Donec et dui non massa tristique adipiscing.